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GENERAL THEORY AND METHODS FOR MATRIC FACTORING

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Methods are developed for factoring an arbitrary rectangular matrix S of rank r into the form FP, where F has r columns and P has r rows. For the statistical problem of factor analysis, S may be the score matrix of a population of individuals on a battery of tests. Then F is a matrix of factor loadings, P is a matrix of factor scores, and r is the number of factor variates. (As in current procedures, there remains a subsequent problem of rotation of axes and interpretation of factors, which is not discussed here.) Methods are also developed for factoring an arbitrary Gramian matrix G of rank r into the form FF', where F has r columns and F' denotes F transposed. For the statistical problem of factor analysis, G may be the matrix of intercorrelations, R, of a battery of tests, with unity, communalities, or other parameters in the principal diagonal. R is proportional to SS', and it is shown that S can be factored by factoring R. This may usually be the most economical procedure in practice; it should not be overlooked, however, that S can be factored directly. The general methods build up an F (and P) in as many stages as desired; as many factors as may be deemed computationally practical can be extracted at a time. Perhaps it will usually be found convenient to extract not more than three factors at a time. Current procedures, like the centroid and principal axes, are special cases of a general method presented here for extracting one factor at a time.

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1. *Introduction*

This paper develops a variety of new methods for extracting factors from matrices. It gives methods for extracting one factor at a time and methods for extracting several factors at a time, be they oblique or orthogonal. Prevalent methods, like the centroid and principal axes, are shown to be special cases of those developed here.

As an example of methods new to factor analysis, it is shown that "weighted centroid" methods can be used instead of the centroid method, that there is no need to be restricted to ± 1 weights in summing columns in the correlation matrix to determine factor loadings. By judicious choice of weights, according to their particular data, research workers can reduce mean square residuals more rapidly at each stage than by present routines, and without a disproportionate increase in labor.

A variety of new techniques is given for extracting *more* than one factor at a time. It should be feasible to routinize the extraction of two or three factors at a time by the formulas developed below.

Many other possible variations in technique follow from the general formulas established here.

The methods presented are actually adaptations and generalizations of a technique due to Lagrange for reducing bilinear and quadratic forms (**10**, p. 68). They are justified by three theorems established below (see §§3, 7, and 8), one of which is a generalization of that of Lagrange. These three theorems provide a complete theory of matric factoring for factor analysis.

The formulas are derived rigorously. Experience has shown that lack of attention to rigor often leads to unsuspected pitfalls in statistical methods for the social and psychological sciences. Factor analysis in particular has now reached a stage which makes it necessary that the discipline be put on a more rigorous foundation. As an example, in the Spearman-Thurstone approach there are unsolved problems associated with the determination of communalities which—when solved—may conceivably vitiate a considerable amount of published empirical work. In this paper, we shall assume that the problems of communalities can be taken care of, so that the factoring of matrices in the Spearman-Thurstone approach is included.

Attention to rigor has other practical benefits. To my knowledge, no complete justification has been published before for the factoring method most widely in use—Thurstone's centroid method (**9**). It was trying to prove this and related things that led to the uncovering of the more general methods presented here.

Thurstone's centroid method is partly justified—assuming the communality problems are cleared up if communalities are used—as a special case of Lagrange's method. As is shown below, Lagrange's theorem proves that the centroid method does reduce the rank of a Gramian matrix by unity at each stage. Completion of the justification is provided by theorems (2) and (3) of §7 and §8 below.

Because of the practical and theoretical suggestions forthcoming from such a treatment, the theory of factoring is presented first from

a general matric point of view. The case of the factoring of an arbitrary rectangular matrix is discussed first. This then includes the direct factoring of a test score matrix. The methods are seen to apply in particular to Gramian matrices, of which correlation matrices are a special case.

2. *Some Properties of Gramian Matrices*

In the demonstrations to be given below, certain properties of Gramian matrices will be utilized. Proof of these properties can be found elsewhere in a systematic treatment of Gramian matrices (3), so we shall simply list here those theorems assumed in the present paper. The theorems will be stated here only as for real matrices for convenience.* The transpose of a matrix $\boldsymbol{M}$ is denoted by $\boldsymbol{M}'$.

Definition.† A Gramian matrix is a symmetric matrix in which all principal minors of all orders are nonnegative. (3, p. 49)

Theorem (*a*). If $\boldsymbol{M}$ is any rectangular matrix of rank r, then $\boldsymbol{MM}'$ is Gramian and of rank r. (3, p. 53)

Theorem (*b*). For any Gramian matrix $\boldsymbol{G}$ of order $n \times n$ and of rank $r > 0$ there exists an $\boldsymbol{F}$ of order $n \times r$ (and of rank r) such that $\boldsymbol{FF}' = \boldsymbol{G}$. (3, p. 55)

Theorem (*c*). If $\boldsymbol{G}$ is Gramian and nonsingular, then all principal minors of $\boldsymbol{G}$ of all orders are positive. (3, p. 56)

Theorem (*d*). If $\boldsymbol{M}$ is any matrix whose number of columns equals the order of $\boldsymbol{G}$, where $\boldsymbol{G}$ is Gramian, then $\boldsymbol{MGM}'$ is Gramian (3, p. 75).

Theorem (*e*). Every Gramian matrix has a unique Gramian square root. (3, p. 121).

Theorem (*f*). If $\boldsymbol{MM}' = \boldsymbol{FF}' = \boldsymbol{G}$, where $\boldsymbol{F}$ is as defined in theorem (b), then there exists an orthogonal matrix $\boldsymbol{U}$ such that

$$\boldsymbol{M} = \| \boldsymbol{F}\, 0 \| \boldsymbol{U} \qquad (3, \text{ p. } 68).$$

3. *The Generalization of Lagrange's Theorem*

In this and the next section, the problem of factoring will be dealt with from a purely matric point of view, and special treatment will be made later of the case of statistical factors.

We first need to establish the following lemma.

Lemma 1. If S *is of order* $n \times N$ *and of rank* $r > 0$, *then for any in-*

* The treatment in (3) is for general complex Gramian matrices; and actually, the theorems for matric factoring of the present paper hold with but a change in wording for complex matrices.

† An equivalent definition, as can be seen from theorems (a) and (b), is as follows: a matrix expressible in the form $\boldsymbol{MM}'$. Other equivalent definitions are possible.

teger s *such that* $0 < s \leqq r$ *there exists an infinite number of pairs of matrices* X *and* Y *of orders* s×n *and* s×N, *respectively, such that* XSY′ *is nonsingular. In fact, all nonsingular matrices of order* s×s *are expressible in the form* XSY′.

The proof follows from a theorem in linear dependence which states (1, p. 67) that there always exist nonsingular matrices A_n and A_N, of orders $n \times n$ and $N \times N$, respectively, such that

$$A_n S A'_N = \begin{Vmatrix} I_r & 0 \\ 0 & 0 \end{Vmatrix},$$

I_r being the unit matrix of order $r \times r$. Let X_0 be the first s rows of an A_n, and let Y_0 the first s rows of a corresponding A_N. Then $X_0 \underline{S} Y'_0 = I_s$, where I_s is the unit matrix of order $s \times s$. Let B and C be arbitrary nonsingular matrices of order $s \times s$, and let

$$X = BX_0, \qquad Y = CY_0.$$

Then

$$XSY' = BX_0 S Y'_0 C' = \underline{BI_s} C' = BC'.$$

Clearly the last member is nonsingular, so that the first member is nonsingular.

Since A_n and A_N can be determined in infinitely many ways, so can X_0 and Y_0, and all the more so can X and Y. Moreover, since B and C are arbitrary, any nonsingular* matrix of order $s \times s$ can be expressed in the form BC', hence in the form XSY'.

The following theorem is basic to the entire discussion.

Theorem 1. Let S *be any matrix of order* n×N *and of rank* $r > 0$. *Let* X *and* Y *be of orders* s×n *and* s×N, *respectively (where* $s \leqq r$), *and such that* XSY′ *is nonsingular. Then the residual matrix*

$$S_1 = S - SY'(XSY')^{-1}XS \tag{1}$$

is exactly of rank $r - s$.

This theorem generalizes a result due to Lagrange (10, p. 68) which can be stated as follows:

Lagrange's Theorem: Let S *be any matrix of order* $n \times n$ *and of rank* $r > 0$. *Let* x *and* y *be row vectors of* n *elements each and such that* $xSy' \neq 0$. *Then the residual matrix*

$$S_1 = S - \frac{Sy'xS}{xSy'}$$

is exactly of rank $r - 1$.

* More generally, if we do not restrict B and C to being nonsingular, we prove that *any* matrix of order $s \times s$ is expressible in the form XSY'.

For the proof of theorem (1) (which will prove Lagrange's theorem in particular), we first note that lemma (1) assures the existence of an infinite number of satisfactory matrices X and Y. Form the supermatrix

$$M = \begin{Vmatrix} I_s & (XSY')^{-1}XS \\ SY' & S \end{Vmatrix},$$

where I_s is again the unit matrix of order $s \times s$, and M is of order $(n+s) \times (N+s)$. Postmultiplying the first submatric column of M by $(XSY')^{-1}XS$ and subtracting from the second column, and using (1), yields

$$M_1 = \begin{Vmatrix} I_s & 0 \\ \hline SY' & S_1 \end{Vmatrix}.$$

Premultiplying the first row of M_1 by SY' and subtracting from the second row yields

$$M_2 = \begin{Vmatrix} I_s & 0 \\ 0 & S_1 \end{Vmatrix}.$$

Premultiplying the second submatric row of M by $(XSY')^{-1}X$ and subtracting from the first row yields

$$M_3 = \begin{Vmatrix} 0 & 0 \\ SY' & S \end{Vmatrix}.$$

Postmultiplying the second column of M_3 by Y' and subtracting from the first column yields

$$M_4 = \begin{Vmatrix} 0 & 0 \\ 0 & S \end{Vmatrix}.$$

Now M, M_1, M_2, M_3, and M_4 differ from each other only by elementary transformations, so that they all have the same rank. By inspection, the rank of M_2 is the rank of S_1 plus s (the rank of I_s); and the rank of M_4 is r, the rank of S. Therefore, the rank of S_1 is exactly $r - s$.

An important additional fact is that

$$XS_1 = 0, \qquad S_1Y' = 0, \tag{2}$$

as is readily verifiable by premultiplying both members of (1) by X, and by postmultiplying each member by Y'. Therefore, if it is desired to "factor" S_1 further by the method of the theorem just established,

neither X nor Y can be used. The condition that $X_1S_1Y'_1$ be nonsingular, where the matrices with subscripts now are to replace those without subscripts in theorem (1), would not be satisfied if X and/or Y is used for X_1 and/or Y_1.

By choosing X_1 and Y_1 of the appropriate rank and such that $X_1S_1Y'_1$ is nonsingular, the rank of S_1 can be reduced again by any desired amount, leaving a residual matrix S_2. The process can be continued until a residual matrix of zero is obtained.

4. *The Factor Matrices*

A matrix S, of order $n \times N$ and of rank $r > 0$, will be said to be factored by matrices F and P if F and P are of orders $n \times r$ and $r \times N$, respectively, and such that $S = FP$. Then F and P are necessarily of rank r, for their ranks cannot exceed r because of their orders, and the rank of S cannot be greater than either of their ranks since S is their product.

Evidently, if one pair F and P exists, then there is an infinite number of pairs of such factor matrices for a given S. If Z is an arbitrary nonsingular matrix of order $r \times r$, then

$$F_0 = FZ^{-1}, \qquad P_0 = ZP$$

will define a new pair of factor matrices, for F_0 and P_0 are of the requisite rank and orders, and $S = F_0P_0$.

By using theorem (1), we can build up an F and P in stages, incidentally proving that *every* S *can be factored.* In theorem (1), let

$$F_1 = SY'(XSY')^{-1}Z_1^{-1}, \tag{3}$$

where Z_1 is an arbitrary nonsingular matrix of order $s \times s$, and let

$$P_1 = Z_1XS. \tag{4}$$

Then equation (1) can be rewritten as

$$S_1 = S - F_1P_1. \tag{5}$$

Subsequent residual matrices can be similarly treated until, if the mth residual matrix is the first to be zero, we have

$$\begin{aligned} S_2 &= S_1 - F_2P_2 \\ S_3 &= S_2 - F_3P_3 \\ &\cdots\cdots \\ 0 &= S_{m-1} - F_mP_m. \end{aligned}$$

Adding corresponding members of these m equations, we see that

$$S = F_1P_1 + F_2P_2 + \cdots + F_mP_m.$$

Therefore, if we form the supermatrices

$$F = ||F_1 F_2 \cdots F_m||, \qquad P = \left\| \begin{matrix} P_1 \\ P_2 \\ \cdot\ \cdot \\ P_m \end{matrix} \right\|,$$

then F and P will be factor matrices of S, since they are of order $n \times r$ and $r \times N$, respectively, by construction, and $\underline{S} = FP$.

5. *The Problem of Metric*

In the statistical problem of factor analysis, the matrix S to be factored is a matrix of scores of N individuals on n tests. These scores are usually expressed in standard form: with zero means and unit variances. Or else we can consider the case where unique factor scores are subtracted out, so that S represents reduced test scores, with zero means again, but with communalities as variances.

A factor matrix P represents the scores of the N individuals on a set of r (common) factors, and these factor scores are usually desired to be expressed in standard form. The corresponding factor matrix F is then said to contain the loadings of the n tests on the r factors.

We can easily show that if the means of the test scores are all zero, then the means of the factor scores in P_1 are all zero. Form the row vector of N elements, each of which is unity,

$$\Sigma_N = ||1\,1\,1 \cdots 1||.$$

To say that the test scores have zero means is to say that

$$S\Sigma'_N = 0\,. \tag{6}$$

Postmultiplying both members of (4) by Σ'_N, and using (6),

$$P_1\Sigma'_N = 0\,, \tag{7}$$

or the s sets of factor scores have zero means.

Postmultiplying both members of (5) by Σ'_N, and using (6) and (7), we see that the right member becomes zero, so that the means of the rows in S_1 are all zero. Hence, the means of the factor scores in P_2 are zero. Similarly, we see that the means of the rows of all the remaining residual matrices are zero, and the means of all the remaining factor scores are zero.

It is quite another matter, however, to discuss the variances of the factor scores, and even more so the intercorrelations between factors. If the tests are in standard form, then their correlation matrix is

$$\boldsymbol{R}=\frac{1}{N}\boldsymbol{SS}', \tag{8}$$

the diagonal elements being unity. If the tests have unique factor scores subtracted out, then the diagonal elements in (8) are communalities. The covariance matrix of the s factor scores in $\boldsymbol{P}_1$ (which we now know to have zero means) is, according to (4), and using (8),

$$\frac{1}{N}\boldsymbol{P}_1\boldsymbol{P}'_1=\frac{1}{N}\boldsymbol{Z}_1\boldsymbol{XSS}'\boldsymbol{X}'\boldsymbol{Z}'_1=\boldsymbol{Z}_1\boldsymbol{XRX}'\boldsymbol{Z}'_1. \tag{9}$$

In order for these factor scores to have unit variances, the principal diagonal elements of the covariance matrix must all be unity. One way of achieving this—as can be seen from the last member of (9)—would be to let $\boldsymbol{Z}_1$ be a diagonal matrix whose diagonal elements are reciprocals* of the (positive) square roots of the corresponding principal diagonal elements of $\boldsymbol{XRX}'$. Another way would be to modify any given $\boldsymbol{Z}_1$ by premultiplying it by a diagonal matrix whose elements are reciprocals* of the (positive) square roots of the corresponding elements in the principal diagonal of the last member of (9), and using this modified matrix to replace the old $\boldsymbol{Z}_1$.

In order to obtain *orthogonal* or *uncorrelated* factor scores, the left member of (9) would have to be the unit matrix. Therefore, in order to obtain orthogonal factors, $\boldsymbol{Z}_1$ would have to be determined to make the last member of (9) the unit matrix.

A similar discussion holds separately for each of $\boldsymbol{P}_2, \boldsymbol{P}_3, \cdots, \boldsymbol{P}_m$.

There doesn't seem to be much profit in discussing here methods for getting orthogonal factors for the most general case, so this topic is not pursued further here. Instead, we go on to the following special method which is of considerable importance.

6. *Orthogonal Factors*

The following lemma will be needed here.

Lemma 2. If G *is Gramian, of order* n×n, *and of rank* r>0, *then for any integer* s *such that* $0 < s \leqq r$ *there exists an infinite number of matrices* X *of order* s×n *such that* XGX′ *is nonsingular. In fact,*

* The diagonal elements are not zero and are actually positive. To see this, note that the rank of $\boldsymbol{P}_1\boldsymbol{P}'_1$ must be the rank of $\boldsymbol{P}_1$ [theorem (a)], so that the covariance matrix is a nonsingular Gramian matrix. Hence all principal minors must be positive [theorem (c)]; in particular, the principal diagonal elements must be positive. $\boldsymbol{XRX}'$ is also Gramian since $\boldsymbol{R}$ is Gramian [theorem (d)], and is nonsingular since it differs from $\boldsymbol{P}_1\boldsymbol{P}'_1$ only by nonsingular transformations. Hence its diagonal elements are also all positive.

all nonsingular Gramian matrices of order s×s *are expressible in the form* XGX′.

The proof follows from the fact that any symmetric matrix can be diagonalized by an orthogonal matrix (10, p. 90). Thus, there exists an orthogonal matrix U such that

$$UGU' = \begin{Vmatrix} D^2_r & 0 \\ 0 & 0 \end{Vmatrix},$$

where D_r is a diagonal matrix whose diagonal elements are the (positive) square roots of the r positive latent roots of G. [That the nonzero latent roots of G must all be positive is well known. To verify this directly, note that D^2_r must be Gramian, since UGU' must be Gramian by theorem (d).] Form the $n \times n$ matrix

$$A = \begin{Vmatrix} U_r D_r^{-1} & A_1 \\ 0 & A_2 \end{Vmatrix},$$

where U_r is an arbitrary orthogonal matrix of order $r \times r$, A_1 is an arbitrary matrix of order $r \times (n - r)$, and A_2 is an arbitrary matrix of order $(n - r) \times (n - r)$. It can be verified by direct multiplication that

$$A(UGU')A' = \begin{Vmatrix} I_r & 0 \\ 0 & 0 \end{Vmatrix}.$$

Let X_0 be the first s rows of AU. Then $X_0 G X'_0 = I_s$. Let B be an arbitrary nonsingular matrix of order $s \times s$, and let $X = BX_0$. Then

$$XGX' = BX_0GX'_0B' = BI_sB' = BB'.$$

Since the last member is nonsingular, the first must be.

Because of the arbitrary submatrices in A, X_0 can be determined in an infinite number of ways (assuming $n > 1$ if $r = n$), and hence all the more so can X. Since B is arbitrary, any nonsingular* Gramian matrix of order $s \times s$ can be expressed in the form BB' by theorem (b), and hence can be expressed in the form XGX'.

Returning to theorem (1), instead of considering an arbitrary Y, let

$$Y = XS, \tag{10}$$

* More generally, if we do not restrict B to being nonsingular, we prove that *any* Gramian matrix of order $s \times s$ is expressible in the form XGX'.

where X is still arbitrary, except again for the restriction that

$$XSY' = XSS'X' \tag{11}$$

be nonsingular. SS' is Gramian and of rank r by theorem (a), so that the existence of many such matrices X is assured by lemma (2). Using (10) and (11) in (3) and (4), and letting*

$$Z_1 = \sqrt{N}(XSS'X')^{-\frac{1}{2}},$$

we get

$$F_1 = \frac{1}{\sqrt{N}} SS'X'(XSS'X')^{-\frac{1}{2}} \tag{12}$$

$$P_1 = \sqrt{N}(XSS'X')^{-\frac{1}{2}}XS. \tag{13}$$

Then

$$\frac{1}{N} P_1P'_1 = (XSS'X')^{-\frac{1}{2}}XSS'X'(XSS'X')^{-\frac{1}{2}} = I.$$

The last member shows that the factor scores have unit variances and are orthogonal. Therefore (12) and (13) provide formulas for direct computations of orthogonal factors.

Note that, using (8), we can rewrite (12) and (13) as

$$F_1 = RX'(XRX')^{-\frac{1}{2}} \tag{14}$$

$$P_1 = (XRX')^{-\frac{1}{2}}XS. \tag{15}$$

These are more compact formulas for direct computations of orthogonal factors.

Some additional facts of some interest, although not very relevant here, are as follows. Postmultiplying both members of (15) by $\frac{1}{N}S'$, and again using (8), we see that

$$\frac{1}{N} P_1S' = (XRX')^{-\frac{1}{2}}XR,$$

or, comparing the right member with that of (14),

$$\frac{1}{N} P_1S' = F'_1. \tag{16}$$

Equation (16) states that the factor loadings are the *covariances* between tests and factors.

* The Gramian square root of $XSS'X'$ could be used, for example [theorem (e)].

An additional bit of information can be got by premultiplying (14) by X, which yields

$$XF_1 = (XRX')^{-1} = F'_1X' . \tag{17}$$

Formulas similar to (12) and (13) hold for $F_2, P_2, \cdots, F_m, P_m$, yielding factors which are orthogonal *within* each set. However, with no further restrictions on the X_j, the factors will not in general be orthogonal *between* sets. That is, in general, if $j \neq k$,

$$P_jP'_k \neq 0 ,$$

whereas if $j = k$,

$$\frac{1}{N} P_jP'_j = I.$$

In the computational methods based on the correlation matrix, to be described next, factors will be orthogonal between as well as within sets.

7. *The Justification for Computations Based on the Correlation Matrix*

Current techniques for factoring operate on the correlation matrix R rather than on the score matrix S. The usual purpose is to find an F of order $n \times r$ and of rank r such that $R = FF'$.

As we shall see, this implies orthogonal factor scores. In order to justify such a procedure, it is necessary to prove the following theorem:

Theorem 2. If

$$R = \frac{1}{N} SS' = FF' , \tag{18}$$

where S *is of order* n $\times$ N *and of rank* r, *and* F *is of order* n $\times$ r (*and of rank* r), *then it is possible to determine a* P *of order* r $\times$ N *and such that*

$$S = FP. \tag{19}$$

Furthermore, this P *is uniquely determined, and it satisfies*

$$\frac{1}{N} PP' = I. \tag{20}$$

The existence of a P satisfying (19) follows from theorem (f), setting $\frac{1}{\sqrt{N}} S = M$, and letting $\frac{1}{\sqrt{N}} P$ be the first r rows of U. Once

the existence of such a P is established,* the fact that it is unique can be immediately shown by premultiplying both members of (19) by† $(F'F)^{-1}F'$, yielding

$$P = (F'F)^{-1}F'S\,. \tag{21}$$

The right member of (21) is clearly unique for a given F.

That P satisfies (20) follows from the fact that U is orthogonal. Actually, we can postmultiply each member of (21) by its transpose, divide by N, and use (18):

$$\frac{1}{N}PP' = \frac{1}{N}(F'F)^{-1}F'SS'F(F'F)^{-1} = (F'F)^{-1}F'FF'F(F'F)^{-1} = I.$$

This is direct verification that P satisfies (20).

This last result proves that no matter how the F is built up, the corresponding factor scores are automatically orthogonal as long as $R = FF'$. Hence, if such an F is built up by subtests, as is done in the following remaining pages, the corresponding factor scores are automatically orthogonal both within and between subsets.

8. *Factoring the Correlation Matrix*

Current techniques, like Thurstone's centroid method, usually build up a matrix F by computing one column at a time. In the terminology of factor analysis, one factor is extracted at a time.

From theorem (1), we can get a method for extracting *as many factors as we wish at a time*, operating directly on the correlation matrix. Since this will be a general method for factoring any Gramian matrix, let us express it in general terms.

The fundamental theorem for the method is as follows:

Theorem 3. Let G *be a Gramian matrix of order* n×n *and of rank* r > 0. *Let* X *be of order* s×n *and such that* XGX′ *is nonsingular. Then the residual matrix*

$$G_1 = G - GX'(XGX')^{-1}XG \tag{22}$$

is of rank r − s *and is Gramian.*

The proof of the first half of the theorem, that the rank of G_1 is $r - s$, is given by theorem (1), substituting G for S and letting $Y = X$. Lemma (2) assures the existence of infinitely many satisfactory matrices X.

* Holzinger and Harman (4, pp. 289-290) tackle a problem similar to that of theorem (2) in attempting to show the relations between sets of factors. However, what they actually accomplish is to provide a formula, resembling (21) above, for transforming one solution into another *if the transformation is possible.* But the problem is to show that the transformation is possible. Proof of this follows from theorem (f).

† $F'F$ is nonsingular since its rank must be that of F [theorem (a)].

To prove the second half, that G_1 is Gramian, let $G = F_0F'_0$, where F_0 is of order $n \times r$, according to theorem (b). Then (22) can be rewritten as

$$\begin{aligned} G_1 &= F_0F'_0 - F_0F'_0X'(XF_0F'_0X')^{-1}XF_0F'_0 \\ &= F_0[I - F'_0X'(XF_0F'_0X')^{-1}XF_0]F'_0. \end{aligned}$$

In the last member, denote the symmetric matrix in brackets by H. It is readily verified by direct multiplication that $H = H^2$, or we can write

$$H = HH'. \tag{23}$$

From theorem (a), the right member of (23) is Gramian, so that H is Gramian. Since $G_1 = F_0HF'_0$, G_1 must also be Gramian from theorem (d).

To obtain s columns of a factor matrix F for G, we can let

$$F_1 = GX'(XGX')^{-\frac{1}{2}},$$

which is of order $n \times s$, and where the Gramian square root, say, of XGX' is used. Then (22) can be rewritten as

$$G_1 = G - F_1F'_1.$$

Since G_1 is Gramian, the factoring process can be reapplied to G_1, and similarly to subsequent residual matrices, until the remaining $r - s$ columns of an F are established.

As a special case of equation (2), we note that

$$\underline{XG_1} = 0, \tag{24}$$

so that new X_j must be used each time for each subset.

For the case of statistical factors, the Gramian matrix G is the correlation matrix R, with usually either unity or communalities in in the diagonal.

9. *The Special Case of the Centroid Method*

The calculation of $(XGX')^{-\frac{1}{2}}$ may in general constitute the major obstacle to successful application of the preceding method when $s > 3$. It should not be difficult to work out relatively simple procedures for the case $s \leqq 3$.

In the special case where $s = 1$, the work is extremely simple. X becomes simply a row vector, say x, and xGx' is simply a positive scalar number. Hence $(xGx')^{-\frac{1}{2}}$ is easily computed by ordinary arithmetic. In fact, we can write F_1 as a column vector, say

$$f_1 = \frac{Gx'}{\sqrt{xGx'}}. \tag{25}$$

In the special case where the elements of x are chosen so as to be all unity, say

$$x = \Sigma_n = ||1\ 1 \cdots 1||,$$

we have *Thurstone's centroid method,* where

$$f_1 = \frac{G\Sigma'_n}{\sqrt{\Sigma_n G \Sigma'_n}}.$$

As a special case of (24), we have $\Sigma_n G_1 = 0$, so that Σ_n cannot be used for subsequent factoring, as is well known in Thurstone's method.

In Thurstone's method, reflections of axes are made for the extraction of subsequent factors. This simply means that for factoring the residual matrices, vectors of the type

$$x_j = ||\pm 1\ \pm 1\ \pm 1 \cdots \pm 1||$$

are used, where the signs may be chosen in any manner, just so $x_j G_j x'_j \neq 0$ (in which case $x_j G_j x'_j > 0$, since G_j is nonnegative definite).

The present analysis shows that *there is no need to be restricted to* x_j *whose elements are* ± 1. Especially if Thurstone's centroid is used for the first factor loadings, it seems that the elements of subsequent residual matrices will in general tend to diminish more rapidly in the sense of least squares by judicious choice of x_j with an eye to maximizing $x_j G_j x'_j / x_j x_j'$ (cf. 8).

For the statistical problem of factor analysis, further investigation of this method should yield dividends in practical procedures that will reduce mean square residuals rapidly at each stage, without involving the labor of computing principal axes. An alternative goal would be to try to arrive as closely as possible to a simple structure —if it exists— in the process of extracting the factors, so as to minimize a subsequent problem of rotation.

10. *The Special Case of Principal Axes*

As is well known, the method of principal axes minimizes the mean square residual at each stage (6, 8). We shall develop this method here as a special case of Lagrange's method.

Returning to the matrix S and §6, consider the case where $s = 1$, so that X and Y are row vectors, say x and y. Suppose that x and y are chosen to satisfy the simultaneous matric equations

$$xS = \lambda y \tag{26}$$

$$yS' = \mu x, \tag{27}$$

where λ and μ are scalars to be determined. Substituting $1/\lambda$ times the left member of (26) for y in (27), and multiplying through by λ,

$$xSS' = \lambda\mu x . \tag{28}$$

This shows that x is a latent vector of SS' (and hence of $\boldsymbol{R}$), and $\lambda\mu$ is a latent root of SS'.

Similarly, by substituting $1/\mu$ times the left member of (27) for x in (26), and multiplying through by μ,

$$yS'S = \lambda\mu y . \tag{29}$$

Hence* y is a latent vector of $S'S$ corresponding to the same latent root $\lambda\mu$ as does x.

Equations (26) and (27) can be used as a basis for an iterative method for principal axes that operates on the *test scores* rather than on the correlation matrix. This would be of advantage, for instance, when the intercorrelations between the tests are not desired, but only the major principal axis is sought. In such a case, it may be less work to perform iterations directly on the scores rather than to compute all the intercorrelations first in order to perform iterations on $\boldsymbol{R}$.

Letting x_k, y_k, λ_k, and μ_k be values of the trial vectors and constants of proportionality at the kth trial, the iterations take the form

$$\lambda_k y_k = x_k S , \quad \mu_{k+1} x_{k+1} = y_k S' .$$

Proof that convergence will take place in general to the major axis follows immediately from the proof for iterations on the correlation matrix (cf. 6, 8), as inspection of the transition from equations (26) and (27) to (28) and (29) will show.

When $s > 1$, iterations on the score matrix may not be too feasible, for the general case, to get several principal axes simultaneously.

Returning to the factoring of a Gramian matrix—like the correlation matrix— when $s = 1$, we see that a principal axis is a special case of (25), where now x satisfies an equation

$$x\underline{G} = \xi^2 x ,$$

ξ^2 being a latent root of $\boldsymbol{G}$. If x is normalized,

$$x\boldsymbol{G}x' = \xi^2 xx' = \xi^2 ,$$

* We have tacitly assumed that $\lambda\mu \neq 0$. (28) and (29) obviously are true if $\lambda\mu = 0$. There will be $n-r$ independent solutions to (28) for $\lambda\mu = 0$, and there will be $N-r$ for (29). All such solutions, however, are of no interest for the present factor problem, since only those r solutions corresponding to the r positive latent roots go into the factor matrices.

so that (25) can be written in this case as

$$f_1 = \xi x' .$$

Hotelling has discussed iterations on the correlation matrix for the case $s = 1$ (6, 7).* Duncan, Frazer, and Collar (2) have discussed this type of iteration for a more general matrix. Horst (5) has generalized iterations on the correlation matrix for the case $s > 1$, and proof for this generalization has been provided by Householder and Young (8).

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* See also his recent paper, Some new methods in matrix calculation, *Annals Math. Statistics,* 1943, 14, 1-34, especially part III.